



OPERATION Department
SIMHADRI

Note No :1



Ref:SMTTP/OPERATION/2025-26/CYCLONE/416080

Date:10/09/2025(21:20:46)

Subject:STATION PREPAREDNESS FOR CYCLONE

LMI: STATION PREPAREDNESS FOR CYCLONE

Submitted by:

कृष्ण मोहन मारपिना - उप महाप्रबंधक

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On:10/09/2025(21:20:46) **Ref:**SMTTP/OPERATION/2025-26/CYCLONE/416080

Note No:2

Submitted by:

सोमनाथ भट्टाचार्य - अपर महाप्रबंधक

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On:12/09/2025(10:02:38) **Ref:**SMTTP/OPERATION/2025-26/CYCLONE/416080

Note No:3

LMI is in line with the OGN. Forwarded for further review.

Submitted by:

नवीन किशोर - अपर महाप्रबंधक

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On:17/09/2025(12:33:42) **Ref:**SMTPP/OPERATION/2025-26/CYCLONE/416080

Note No:4

Please review page no 19 as discussed.

Submitted by:

राजेश चावला - अपर महाप्रबंधक

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On:17/09/2025(16:56:51) **Ref:**SMTPP/OPERATION/2025-26/CYCLONE/416080

Note No:5

As per HOD-AHM feedback, page number-19 reviewed and following changes incorporated.

Existing: Head of Ash dyke management: Two executives from ADM.

Modified as: Head of Ash dyke management: Two executives from ADM / **AHM**

Modified document is attached as "MODIFIED LMI CYCLONE MANAGEMENT"

Submitted by:

कृष्ण मोहन मारपिना - उप महाप्रबंधक

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On:18/09/2025(21:38:37) **Ref:**SMTPP/OPERATION/2025-26/CYCLONE/416080

Note No:6

May be processed further.

Submitted by:

राजेश चावला - अपर महाप्रबंधक

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On:20/09/2025(15:52:29) **Ref:**SMTPP/OPERATION/2025-26/CYCLONE/416080

Note No:7

Submitted by:

आन्नप्रगद रमा कोटेश्वरराव - अपर महाप्रबंधक

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On:24/09/2025(09:29:43) **Ref:**SMTPP/OPERATION/2025-26/CYCLONE/416080

Note No:8

LMI reviewed and preparedness as per LMI is noted and being adhered .

Forwarded for further approval please.

Submitted by:

टी वी राजशेखर - अपर महाप्रबंधक

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On:26/09/2025(11:52:35) **Ref:**SMTPP/OPERATION/2025-26/CYCLONE/416080

Note No:9

As per the recent development of making ADM dept along with civil employees, responsibility for the points (point no-23 in page no:16 of 27 & point no-ii in page no:22 of 27 for ash pond) related to ash dyke area may please be mentioned as HOD(ADM) instead of HOD(Civil). Rest civil points will remain same as HOD(Civil).

Submitted by:

पी गंगाधर - अपर महाप्रबंधक

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On:30/09/2025(19:43:55) **Ref:**SMTTP/OPERATION/2025-26/CYCLONE/416080

Note No:10

As per HOD-Civil feedback, point number 23 in page number 16 of 27 reviewed and following changes incorporated.

1) Existing responsibility: HOD (Civil) Modified responsibility: HOD (ADM).

2) point number-ii, 22 of 27 page, responsibility allocation: Head of O&M-Civil for Reservoir, Head of ADM for Ash pond.

Modified document is attached as "**MODIFIED LMI CYCLONE MANAGEMENT DATED 14102025**"

Submitted by:

कृष्ण मोहन मारपिना - उप महाप्रबंधक

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On:14/10/2025(06:38:20) **Ref:**SMTTP/OPERATION/2025-26/CYCLONE/416080

Note No:11

Forwarded for further approval please.

Submitted by:

पी गंगाधर - अपर महाप्रबंधक

Padala Hari Veera Gangadhar - 008164 (ADDL. GENERAL MANAGER ,CIVIL MAINT.)
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On:15/10/2025(10:03:43) **Ref:**SMTTP/OPERATION/2025-26/CYCLONE/416080

Note No:12

Reviewed and no change in EMD scope. May please be approved.

Submitted by:

मलय कान्ति जाना - अपर महाप्रबंधक

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On:15/10/2025(15:20:14) **Ref:**SMTTP/OPERATION/2025-26/CYCLONE/416080

Note No:13

Review

Submitted by:

धर्मेन्द्र कुमार तोमर - अपर महाप्रबंधक

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On:16/10/2025(09:42:48) **Ref:**SMTTP/OPERATION/2025-26/CYCLONE/416080

Note No:14

LMI reviewed and needful has been noted for future reference.

May kindly be processed further for needful approval please.

Submitted by:

चिन्मय साहू - उप महाप्रबंधक

CHINMAY SAHU - 100783 (DY. GENERAL MANAGER ,P & S)
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On:22/10/2025(10:01:39) **Ref:**SMTTP/OPERATION/2025-26/CYCLONE/416080

Note No:15

Submitted by:

रोहित प्रकाश - अपर महाप्रबंधक

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On:22/10/2025(10:09:57) **Ref:**SMTPP/OPERATION/2025-26/CYCLONE/416080

Note No:16

Submitted by:

राजेश खैरवा - उप महाप्रबंधक

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On:27/10/2025(12:39:30) **Ref:**SMTPP/OPERATION/2025-26/CYCLONE/416080

Note No:17

Reviewed and may kindly be processed further for needful approval please.

Submitted by:

अनिल कुमार मौर्या - अपर महाप्रबंधक

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On:27/10/2025(15:20:40) **Ref:**SMTPP/OPERATION/2025-26/CYCLONE/416080

Note No:18

Reviewed and forwarded for approval

Submitted by:

नीलांजन मंडल - अपर महाप्रबंधक

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On:27/10/2025(16:11:28) **Ref:**SMTTP/OPERATION/2025-26/CYCLONE/416080

Note No:19

Submitted by:

सुप्रतिम कुरी - उप महाप्रबंधक

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Note No:20

Submitted by:

भरत कुमार राठौर - अपर महाप्रबंधक

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On:27/10/2025(17:59:54) **Ref:**SMTTP/OPERATION/2025-26/CYCLONE/416080

Note No:21

Reviewed and forwarded for approval

Submitted by:

दिव्य चावला - अपर महाप्रबंधक

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On:28/10/2025(09:38:25) **Ref:**SMTTP/OPERATION/2025-26/CYCLONE/416080

Note No:22

Submitted by:

जे आर वर्गीस - अपर महाप्रबंधक

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On:28/10/2025(10:05:29) **Ref:**SMTTP/OPERATION/2025-26/CYCLONE/416080

Note No:23

Reviewed.

Submitted by:

कुम्भारे ममता बी - अपर महाप्रबंधक

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On:29/10/2025(10:24:07) **Ref:**SMTTP/OPERATION/2025-26/CYCLONE/416080

Note No:24

Submitted by:

भक्त बत्सल पात्रा - अपर महाप्रबंधक

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On:29/10/2025(11:10:13) **Ref:**SMTPP/OPERATION/2025-26/CYCLONE/416080

Note No:25

LMI is in line with the OGN. May please be approved.

Submitted by:

नवीन किशोर - अपर महाप्रबंधक

Navin Kishore - 006964 (ADDL. GENERAL MANAGER ,EEMG)
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On:29/10/2025(11:47:42) **Ref:**SMTPP/OPERATION/2025-26/CYCLONE/416080

Note No:26

May please approve.

Submitted by:

हेमेन्द्र कुमार वर्मा - महाप्रबंधक

Hemendra Kumar Verma - 006459 (GENERAL MANAGER ,MAINTENANCE)
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On:30/10/2025(09:30:11) **Ref:**SMTPP/OPERATION/2025-26/CYCLONE/416080

Note No:27

Submitted by:

अहमद कौनैन रज़ा - महाप्रबंधक

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On:30/10/2025(09:55:13) **Ref:**SMTTP/OPERATION/2025-26/CYCLONE/416080

Note No:28

Returned as discussed.

Submitted by:

एस पद्मप्रिय - महाप्रबंधक

S Padmapriya - 004521 (GENERAL MANAGER ,TECH SERVICES)
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On:31/10/2025(12:19:22) **Ref:**SMTTP/OPERATION/2025-26/CYCLONE/416080

Note No:29

Please review as discussed.

Submitted by:

सोमनाथ भट्टाचार्य - अपर महाप्रबंधक

Somnath Bhattacharya - 007203 (ADDL. GENERAL MANAGER ,OPERATION)
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On:01/11/2025(21:04:25) **Ref:**SMTTP/OPERATION/2025-26/CYCLONE/416080

Note No:30

The LMI has been reviewed and forwarded for approval.

Submitted by:

सतीश कुमार मुरपाक - वरिष्ठ प्रबंधक

SATISH KUMAR MURAPAKA - 101773 (SENIOR MANAGER ,OPERATION)
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On:11/11/2025(17:42:26) **Ref:**SMTTP/OPERATION/2025-26/CYCLONE/416080

Note No:31

Reviewed.

May please approve.

Submitted by:

सोमनाथ भट्टाचार्य - अपर महाप्रबंधक

Somnath Bhattacharya - 007203 (ADDL. GENERAL MANAGER ,OPERATION)
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On:12/11/2025(17:38:47) **Ref:**SMTTP/OPERATION/2025-26/CYCLONE/416080

Note No:32

Before putting up LMI for approval, final routing should be through HOD(EEMG), GM(O&M)- to be followed in all cases.

Submitted by:

समीर शर्मा - कार्यकारी निदेशक

Sameer Sharma - 005041 (EXECUTIVE DIRECTOR ,HEAD OF PROJECT)
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On:12/11/2025(18:06:16) **Ref:**SMTTP/OPERATION/2025-26/CYCLONE/416080

Note No:33

LMI is in line with the OGN. May please be approved.

Submitted by:

नवीन किशोर - अपर महाप्रबंधक

Navin Kishore - 006964 (ADDL. GENERAL MANAGER ,EEMG)
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On:12/11/2025(18:47:22) **Ref:**SMTTP/OPERATION/2025-26/CYCLONE/416080

Note No:34

Submitted by:

अहमद कौनैन रज़ा - महाप्रबंधक

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On:13/11/2025(11:13:53) **Ref:**SMTTP/OPERATION/2025-26/CYCLONE/416080

Note No:35

Approved & Submitted by:

समीर शर्मा - कार्यकारी निदेशक

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On:13/11/2025(11:27:40) **Ref:**SMTPP/OPERATION/2025-26/CYCLONE/416080

NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

STATION PREPAREDNESS FOR CYCLONE

LMI/OGN/OPS/SYST/030 Rev: 03

SEPT - 2025

Approved for Implementation by HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	LMI NAME	STATION PREPAREDNESS FOR CYCLONE
2	LMI No.	LMI/OGN/OPS/SYST/030 Rev: 03
3	Reference Document	COS-ISO-00-OGN/OPS/SYST/030 Rev: 03
4	Superseded Document	LMI/OGN/OPS/SYST/030 Rev: 02 May 2022
5	Prepared By	Name: Krishna Mohan Marpina, D G M (Operation)
6	Checked By	Name: Somnath Bhattacharya AGM (Operation)
7	Checked By	Name: Navin Kishore AGM (EEMG)
8	Checked By	Name: Rajesh Chawla AGM (AHM)
9	Checked By	Name: A R Koteswara Rao AGM (CHP)
10	Checked By	Name: T V Rajasekhar AGM (OFS MAINT)
11	Checked By	Name: P H V Gangadhar AGM (Civil MAINT)
12	Checked By	Name: Hemendra Kumar Verma GM (MAINTENANCE)
13	Reviewed By	Name: Ahmad Kaunain Raza G M (O&M)
14	Approved By	Name: Sameer Sharma H O P (Simhadri)

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TITLE	STATION PREPAREDNESS FOR CYCLONE
LMI No.	LMI/OGN/OPS/SYST/030 Rev: 03

1.0 Introduction

In recent past, numbers of cyclones have resulted in machine Non-availability and loss of property in NTPC's coastal area plants like Simhadri and Vallur.

In Oct'14, Hud Hud cyclone at Simhadri plant caused severe damage to transmission lines leading to no power in the township / adjoining area for more than 48 hour. Several teams were formed to accelerate the revival work in the plant and township area. During the recent cyclone on 12.12.16 at Vallur, all units tripped due to transmission line disturbance.

Unit#3 TG barring gear got stalled. Hydrogen and CO2 cylinder were not available in adequate quantity. One Stacker declaimer was not available and there was no coal near the other S/R. Coal feeding was disrupted.

It was observed that many system failures could have been avoided or restoration time could have been reduced by proper mobilization of resources and manpower. Based on learning's during recent cyclones, LMI instructions are being prepared to counter cyclone related issues and emergencies. This will help in ensuring system revival in minimum possible time vis a vis machine availability.

2.0 Objective and Scope

The objective of this LMI is to ensure station preparedness to effectively deal with cyclone. These instructions recognize that problems related to cyclone can be tackled with better preparedness to minimize losses and faster stabilization of units, in case of tripping. This will also help in reducing system revival time in case of emergency situations like power supply failure, non-availability of DG set due to shortage of oil stock etc.

3.0 Superseded Document.

LMI/OGN/OPS/SYST/030 Issue no: 1, Rev no: 2, Dt: May 2022.

4.0 Details of revision

Following Changes are incorporated. In clause 5, description in S No 1, 9, 14, 23, and clause 7 are revised.

4.1 Reference Document:

TITLE	STATION PREPAREDNESS FOR CYCLONE
LMI No.	LMI/OGN/OPS/SYST/030 Rev: 03

- a) This LMI is based on OGN on the subject "Station Preparedness for Cyclone" COS-ISO-00-OGN/OPS/SYST/030 Rev.No:2 Aug 2022
- b) SMTTP Disaster Management Manual, Rev-09, 2022

5.0 Handling Teams & Responsibility

For smooth cyclone management and information flow, following guidelines shall be followed:

- A station level Cyclone Management Group (CMG) shall be identified. Responsibility: HOP
- Head of O&M will be the coordinator/nodal officer for management of Cyclone.
- Apart from O&M members, IT,HR, Medical, CISF and canteen representative shall be the member of this team.
- Group coordinator will also be responsible for all information flow to RED / ROS and ED (OS)/ COS.
- This group will start working after receiving first intimation about expected arrival date and time.
- Head of Operation at COS will be the nodal point to give primary information report about the crisis to ED (OS) / Director (O).
- Station Coordinator / Nodal officer concerned GM-OS shall be responsible for all the thunderstorm related activities of the station.
- ED-OS shall identify the persons to be present in control room at COS depending on type of the emergency.

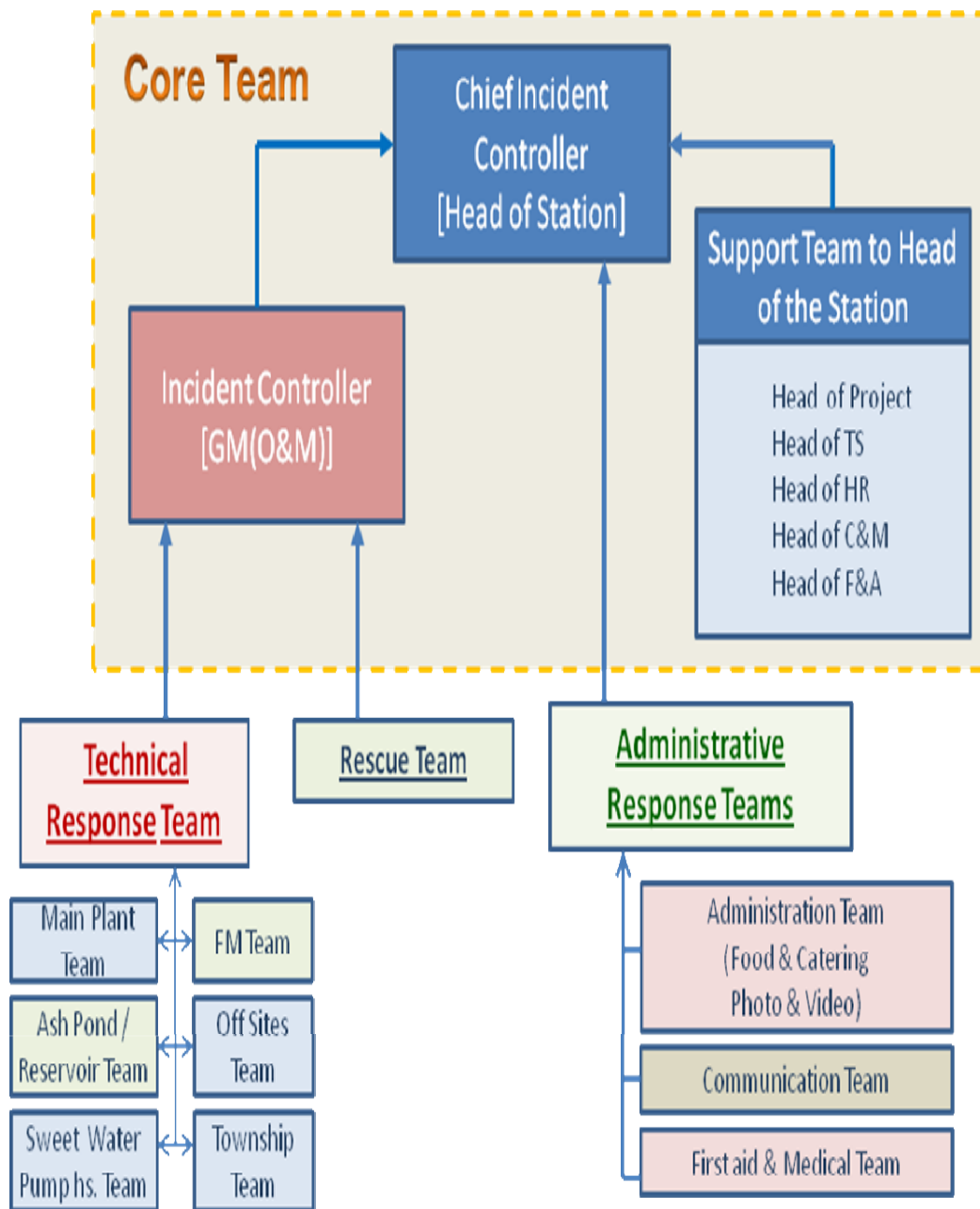
5.1 Cyclone management group

- Coordinator / Nodal Officer: HOD (O&M)
- Members: HOD (O), HOD (Mnt), HOD(EM), HOD(C&I), HOD(civil),HOD (FM),HOD(IT), HOHR,CMO, DC(CISF), Canteen committee chairman.

6.0 Key persons and Their Responsibility

TITLE	STATION PREPAREDNESS FOR CYCLONE
LMI No.	LMI/OGN/OPS/SYST/030 Rev: 03

Various teams have been identified and their roles & responsibilities are explained in the Disaster Management action plan as per the **Annexure I**. The following organization chart illustrates the reporting system in case of cyclone emergency.



TITLE	STATION PREPAREDNESS FOR CYCLONE
LMI No.	LMI/OGN/OPS/SYST/030 Rev: 03

7.0 Suggested enablers for effective operation of Cyclone Management Group

- Availability of regional power map
- Availability of details on alternate source of power in case of grid failure, startup power scheme
- Availability of latest grid normalization procedure
- Availability of contact nos. of all identified executives at station, RHQs, CC, RLDCs and other important persons
- Availability of backup power for control room
- Availability of Man Lifter to remove any loose cladding sheets in boiler and duct area post cyclone
- Availability of web addresses to monitor cyclone movement, wind speed etc.
(https://mausam.imd.gov.in//imd_latest/contents/cyclone.php
<http://www.rsmcnewdelhi.imd.gov.in/https://severeweather.wmo.int/TCFW/>
<https://www.ncmrwf.gov.in/>

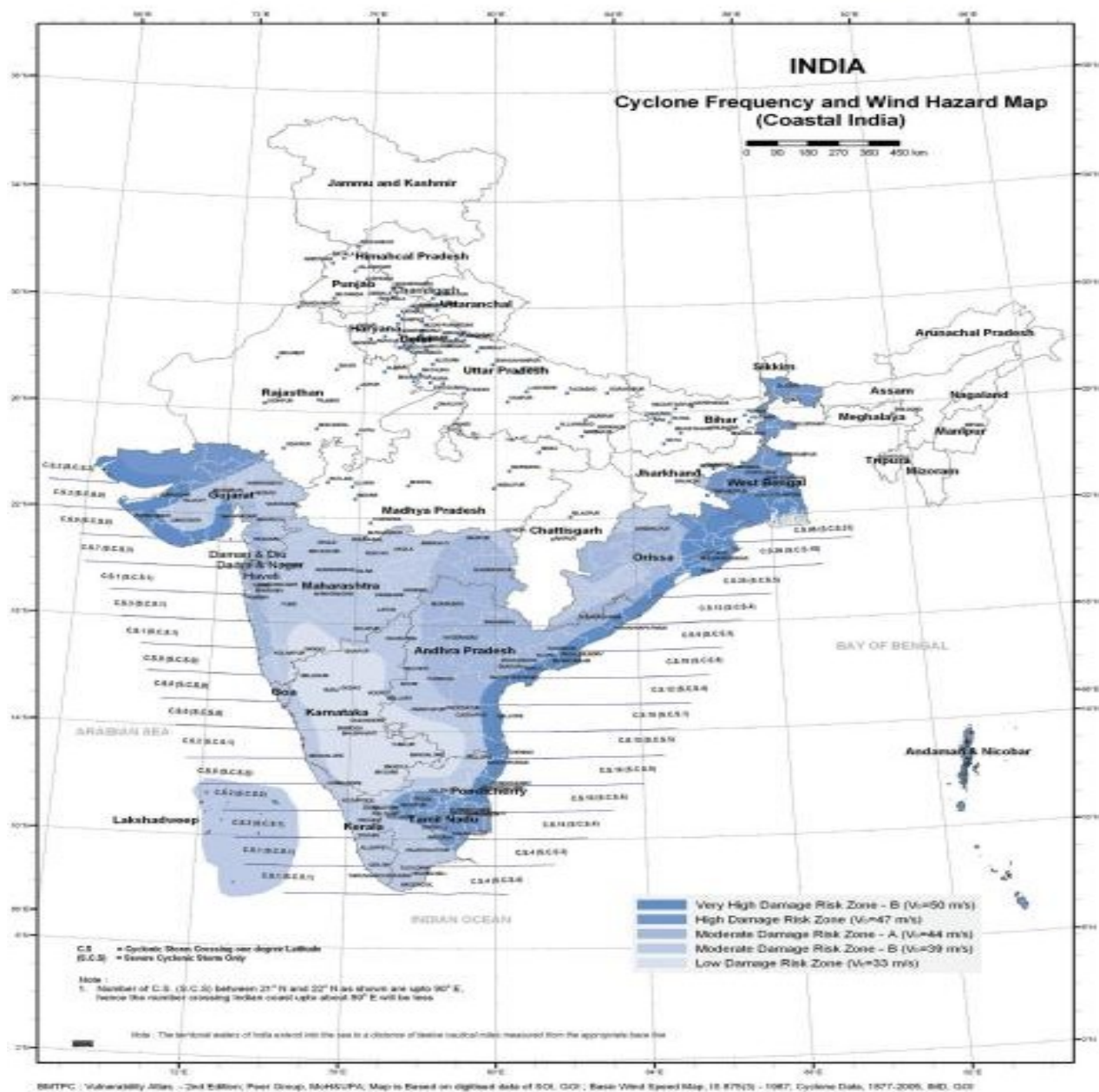
Grading of Cyclone:

Category	Wind speeds
Five	≥70 m/s, ≥137 knots ≥157 mph, ≥252 km/h
Four	58–70 m/s, 113–136 knots 130–156 mph, 209–251 km/h
Three	50–58 m/s, 96–112 knots 111–129 mph, 178–208 km/h
Two	43–49 m/s, 83–95 knots 96–110 mph, 154–177 km/h
One	33–42 m/s, 64–82 knots 74–95 mph, 119–153 km/h

TITLE	STATION PREPAREDNESS FOR CYCLONE
LMI No.	LMI/OGN/OPS/SYST/030 Rev: 03

Cyclone Frequency and Wind Hazard Map as per National Disaster Management Authority Government of India "Cyclone Management " guidelines

Figure 1 (a): Cyclone Frequency and Wind Hazard Map



TITLE	STATION PREPAREDNESS FOR CYCLONE
LMI No.	LMI/OGN/OPS/SYST/030 Rev: 03

8.0 Format for information flow to Corporate Center

8.1 If any Format for MIS on expected Cyclone.

Name of the Station -

Expected date and time of the cyclone.

Sl. No.	Expected date/time of occurrence	Expected Area affected

8.2 Format for MIS after Cyclone

Name of the Station -

Date and time of the Cyclone -

Sl. No.	Type of event	Areas affected	Extent of damage	Expected normalization time

9.0 Records:

9.1.1 Desk Engineer/SCE Log Books

9.1.2 Records of MIS format to Corporate center

10.0 Responsibility

HOD (O) is responsible for implementation of this LMI.

11.0 Review: HOD(O) shall be responsible for reviewing this document on a 2-yearly

Basis or as necessary.

TITLE	STATION PREPAREDNESS FOR CYCLONE
LMI No.	LMI/OGN/OPS/SYST/030 Rev: 03

12.0 Checklist:

S No.	Activity	Details of activity	Responsibility
1.	Sending SMS by Cyclone Management Group Coordinator (Head of O&M)	- To all Cyclone Management Group members, ROS, COS. - To all employees - to store drinking water & candles - to remain indoors to avoid injury due to falling sheets, glass pieces etc during the high winds and alert other persons to close tightly all doors/windows in all offices/buildings to minimize loss to all documents. - Township Administration department also need to be alerted in advance of taking necessary measure to allow save passage of Movement in township and emergency lighting back up in common areas.	HOD(IT)/ Cyclone Management Group members.
2	Communication	- Satellite phone/ other independent communication systems to be made ready. - Walkie-Talkies to be made ready with battery chargers. - Backup Power Supply from DG Set to telephone exchange. - PA system at different places in plant - healthiness to be ensured.	HOD(IT)/ HOD (C&I)
3	Cyclone Management Control Room	Control Room to be set up at appropriate location and work in close association with SCE.	Cyclone Management Group / SCE

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4	Follow District Administration Instructions.	• Cyclone Management Group Coordinator to develop contacts with district officials. • Ensure compliance of their instruction before and during the cyclone.	HOHR/ CMG Coordinator
5	Readiness of all DG sets	• Trial run in every shift after receipt of cyclone information. • All Diesel Tanks to be made full. • Diesel availability for 24 hour DG set running other than the dead stock. • One road tanker (16 KL) to be kept as reserve. • One emergency portable DG to be made available. • A permanent DG system, similar to the one in main plant, to be installed in CHP area.	SCE/HOD (OFS-MM) HOD(CHP) HOD(EMD)
6	Availability of Hydrogen and Carbon dioxide cylinders	• Hydrogen cylinder 01 sets per unit and Carbon dioxide. 02 sets per unit to be made available before the cyclone.	HOD(EM) / HOD (OPRN)
7	All Battery operated instruments (Battery operated Trolley, Search lights, UPS etc.)	• Batteries to be in fully charged condition. • Boost Charging of all batteries to be ensured. • Spare set of batteries to be made available in control Room for search light	HOD(EM) / HOD(C&I)
8	Hydra, JCB, Emergency Portable DG Set & Power Saw.	• Availability of all such emergency systems/equipment to be ensured. • Availability of adequate trained manpower along with gumboots, rain coats • Deployment of Additional vehicles for smooth movement of men and material	All concerned Depts.

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9	Dewatering Pumps	• All Dewatering Pumps to be made ready. • All Diesel Dewatering Pumps at various locations e.g. CW Pump Houses, CHP, Sweetwater, AWRC pump etc. to be checked and made ready. • Barricading of entrance of all underground tunnels by sandbags to be ensured • Emergency boats to be made available. • High capacity pumps to be deployed permanently.	All concerned Depts.
10	Emergency Lights /search lights	• Emergency Lights to be made ready at all locations. • Additional torches along with cells can be made available at all operational locations and to maintenance personnel. • Search lights to be made available. • Lighting masts at different locations to be lowered.	All concerned Depts.
11	Availability Condenser diaphragms	• Condenser diaphragms to be made ready for all units	HOD(TMD)
12	Switchyard, FGD, AHP Control Room & Main Unit Control Room	• Sr. operation executives along with electrical, C&I and Turbine group to remain present in control room during the cyclone to handle any emergency situation. • No Work Permits shall be planned for 400 KV System or major auxiliaries. • SCE to remain in touch with Grid officials. • All MCC Doors and windows to be ensured for tight closure.	Operation, EMD,C&I,TMD, SCE & BMD

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13	Disturbance Recorder	Availability to be ensured.	HOD(EM)
14	Stacker Reclaimer / Tripper safety and coal feeding	To be checked and locked before cyclone. • All brakes, rail clamps, wheel grippers/chokers to be checked for efficacy • All the brakes, rail clamps (i.e hydraulic, electro mechanical, thruster, and manual) to be locked. • At least two pairs of wheels at both the direction to be wedged with rails. Provision of keeping wedges near bogie wheels to be ensured. • For SAC and S/R slew, pinions to be wedged with rack. • Handbrakes of wagons/locos to be locked and wheels at both ends to be tied to rails with chains. • Slew structure / boom structure to be properly locked • S/R to be placed in parking area anchored and locked with parking support structure. • In case, parking is not possible, for whatsoever reason, S/R boom to be lowered and at least one	HOD (CHP)

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		bucket is completely submerged in coal heap Central locking device for slew drive, if not available, needs to be installed. Anemometer should be functional in Stacker Reclaimers. Sufficient coal should remain available within reclaimable range of all S/R All peripheral and Storm water drains should be cleared off any obstruction and Connected to main Channel.	
15	CHP Precautions	• Bunker levels in running units to be made full before cyclone • All conveyors and equipment to be stopped before cyclone. • All gates in the drain network to be lifted to allow free flow of storm water. • Availability of Hydra and JCBs along with operator to be ensured. • Coal unloading at TH/WT to be stopped All persons to be asked to come out of conveyor gallery/ TPs/ Crusher house/Track hopper/S/R/WT/Tunnels etc.	HOD (CHP)
16	Enough Cash to be kept in Hand in different dept	For ready purchase.	All Concerned Depts/ HR
17	Medical Aid	Medical Team along with Ambulance to be on duty round the clock at Main Plant.	CMO
18	Fire & Safety	• Fire and Safety Personnel to be available round the clock at Main Plant. • Boiler lifts to be kept at zero meter. • Boiler Lift Pit dewatering facility to be kept ready	Safety Dept./ CISF / HOD (EM)
19	EOT	Locking of all EOT cranes to be ensured.	HOD(MM)
20	Safety of people	• All structure to be checked for any loose	All concerned Depts. & CMG

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		components (windowpanes /sheets etc) and ensure its proper locking / fitting or removal. • No manpower movement at locations during severe cyclones • A team shall be ready for removal of road & MGR track blockage caused by uprooting of trees and poles with diesel cutters.	
21	Switchyard	• Remove all loose metallic sheets which could cause damage to Electrical equipment's nearby • Remove dead or rotting trees, branches that could fall and cause injury or damage during a cyclone. Secure outdoor objects that could blow away or cause damage • All the outdoor Marshalling Boxes & Junction boxes (isolator, Earth switch, Breaker MB) inside the switchyard should be in locked condition to avoid any spurious outage due to moisture ingress. • Ensure that all the Line protections and Bus bar protections are healthy and in service • In case of any spare CT/CVT/LA/Breakers etc. are available in switchyard, then those spares are to anchored/secured/fasten properly to avoid damage during cyclone. • In case of any ongoing work in the switchyard area, the work is to be suspended and Mobile equipment's (Cranes/Hydra/ Filter Machine etc.), If any are to be removed from the switchyard equipment area to a designated locations and anchor	HOD(EM)
22	Transformer Yard	• Remove all loose metallic sheets which could cause damage to Electrical equipment's nearby	HOD(EM)

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		during a cyclone. • Secure outdoor objects that could blow away or cause damage. Ensure that there are no Loose metallic sheets near the transformer yard • Transformer Oil Filtration Plant should be anchored to Avoid any Movement during Cyclone • Transformer Oil drums (Filled/ empty), if any should be shifted away from the transformer yard area to a secured place. • Incase of any ongoing work in Transformer, the work is to be suspended and associated cranes/ Hydras are to be removed from the working area to a secured place. All the opening of the transformer (removed from the maintenance work) are to be sealed through appropriate blanking plates. • Transformer Marshalling Boxes are to be securely locked.	
23	Ash Dyke	Water level in all the lagoons including OFL(s) to be maintained to minimum possible level. • Free board as envisaged to be maintained so that in the event of sudden rain, there is float available for escape of rainwater through spillways/ WES (water escape structures). • Flexible openings of WES decant wells shall be kept open to the optimum level to allow excess water to escape. • Ensure that all approaches to dyke as well as inspection roads on dykes are easily accessible, free from obstructions (including pipe crossover) and maintained. • Toe drain/ exit to natural drains to be free from obstruction / water stagnation.	HOD(OPRN)/ HOD(AHM)/H OD(ADM)

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		- Sufficient stock of ash filled bags/ sand/ Bottom Ash/ aggregate/ rock builders/ earth/ balla/ empty 50 kg bags etc. to be maintained at probable weak locations under temporary sheds. - Provision for deployment of adequate equipment like excavator / JCB, tipper, tractor, utility vehicle, DG sets, dewatering pumps etc. along with unskilled manpower shall be ensured. - Water logging/ stagnation/ inundation on the downstream side of the dyke to be avoided - Ensure that slopes are protected and the turfing on the downstream side slope is maintained. Concerned dyke O&M team may be equipped with torch, gumboot, umbrella / raincoats etc. to facilitate smooth inspections & maintenance.	
24	Others	- Storage of drinking water in Town ship and plant in sufficient quantity along with pump set arrangement. - Mobile drinking water tankers to be made ready. - Sufficient food stuff stock to be maintained in plant canteen to meet the exigencies - Frequent surveillance of Ash Dyke. - Readiness of 2000 sand/ash bags to meet exigencies. - Locking of loco & wagon at MGR area to be ensured Prior cleanliness of drains to be ensured.	All Concerned Depts / CMG

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Annexure1:**Responsibilities of KEY personnel:****Responsibilities of Chief Incident Controller i.e., Head of the Station:**

Head of the Station has an overall responsibility for directing operations and calling outside help. He will proceed to that Emergency Control Centre and take overall control of the situation. Specific responsibilities/duties and requirements to be ensured by Head of the Station are as under:

- a. Liaison with meteorological department to obtain factual data to assess the gravity of the situation.
- b. Continually review and assess existing and possible developments to determine the most probable course of events and effective methods to deal with them.
- c. Decide in consultation with Works Incident Controller i.e., Head of O&M and direct a safe shut down and evacuation of plant personnel, if required.
- d. Direct and alert the hospital administration for readiness of medical, paramedical and nursing staff and authorise for additional resources, if needed.
- e. Ensure that all employees and their families are informed about the risk and precautions to be taken.
- f. Liaison with District Administration & Police Emergency Authorities.
- g. Ensure accounting of personnel and collate the actual attendance with the master list of persons including contractors and visitors.
- h. Arrange for the rescue / search of missing ones, if any.
- i. Arrange for control of traffic movement within the Plant & township.
- j. Arrange for shifting of material and machinery in the open areas to covered locations. Safe removal of vehicles loaded with flammable or dangerous substances from the plant to safe parking places.
- k. Arrange to maintain chronological record of the events.
- l. Issue authorized statements to the press or the media in consultation with media contact person.
- m. Inform company senior officials from time to time about the developments.

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- n. Control rehabilitation/restoration of affected area after the emergency is over.

Responsibilities of Support Team:

In case of any cyclone emergency warning, they will report to CIC at Emergency Control centre to assist him.

They will -

- Follow all instructions of CIC.
- Maintain a log of events.
- Arrange for urgently required materials through cash purchase or whatever means.
- Arrange funds for various relief measures as well as emergency purchase of materials and sending his representative for emergency purchase.
- Arrange additional vehicles, if any.
- Arrange in advance, alternative communication networks to substitute the existing networks fail.
- Identify suitable staff to act as runners or messengers, if the telephone and other system of communication fail due to any reason.

(iii) Responsibilities of Works Incident Controller, i.e., Head of O&M:

WIC, i.e., Head of O&M operates from plant control room. Heads of MTP and EEMG shall assist him during cyclone emergency. For silent hours, i.e. B and C shifts or on holidays, the Shift In-charge would work in his place until he arrives. The responsibilities of the Works Incident Controller are as under:

- Provide directions / guidelines to the technical response teams and rescue team reporting to him.
- Get under his control the necessary equipment's like cranes, dozers, trucks, vehicles, portable welding and cutting sets, emergency torch lights, battery/diesel operated tree cutting saw sets, emergency lighting towers etc as needed and make available required personnel to operate above facilities.
- Arrange sufficient stock of diesel, petrol etc., for the vehicles engaged in emergency.
- Ensure sufficient number of contractor workers available to do civil jobs like filling sand bags, making bunds, closing/clearing drains and outfalls, excavation etc.
- During Cyclone, assess the scale of emergency.

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- f. Discuss and decide with the Shift In-charge for continuation of plant operations or shutdown. In consultation with CIC, initiate safe shut down of the plant.
- g. Arrange to evacuate the non-essential persons and direct them to report at the Assembly Point.
- h. Set up communication network with the Emergency Control Centre, using Intercom/ walkie- talkie /Mobile phones /alternate communication network.
- i. Assess the external services, if required, and arrange for the same.
- j. Direct all operations within the affected area with following priorities –
 - i. Secure safety of personnel, giving priority to saving life and preventing any injury.
 - ii. Advice and inform as required by the emergency responders, i.e. Fire and Security personnel or emergency services.
- k. Keep CIC informed of the developments from time to time.
- l. Visit to the affected locations to assess the impact and to initiate corrective / control measures immediately after cyclone.
- m. Direct and guide for a safe restoration of plant and facilities.
- n. Make arrangements of temporary lighting / emergency lighting for affected areas, shelters and other places of assembly.
- o. Keep the contact details of all the team members handy, especially for any specific operation vis- a-vis persons, so that they can be contacted when not on duty.

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RESPONSIBILITIES OF TECHNICAL RESPONSE TEAMS:

(i) Main Plant Team:

The team will report to Head of O&M. *List of team members & their responsibilities as follows:*

1. Head of Operations	2. Head of Maintenance	3. Head of Elect. Maint.
4. Head of C&I	5. Head of Fuel Mgmt.	

- a. Identify most likely vulnerable locations and decide whether to continue to operate or not.
- b. Organize safe shutdown of plant, if necessary.
- c. Organize all support service like arranging diesel driven dewatering pumps, alternate communication facilities, commodities like diesel, torch lights, required PPEs etc.
- d. Well before cyclone, the team should inspect all plant drains, storm water drains and outfall for any obstructions and ensure free flow to prevent clogging & flooding of rainwater.
- e. The team shall also inspect the plant locations to identify structures & galleries which need to be strengthened and arrange for necessary preventive actions.
- f. After cyclone, identify the affected plant & equipment and inform Head of O&M and Head of Station.
- g. Plan in consultation with Head of O&M and organize plant restoration process.

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(ii) Ash Pond & Reservoir Team:

The team will report to Head of O&M. *List of team members & their responsibilities as follows:*

1. Head of O&M-Civil for Reservoir	3. Two executives from Civil
2. Head of ADM for Ash pond	4. Two executives from ADM / AHM

- Identify weak & susceptible locations on the bund well before cyclone and initiate strengthening activities.
- Organize for transportation of sand bags, battery operated tower lights and other requisite material to the identified locations.
- Stop all activities before cyclone and arrange for evacuation of people from site to safer locations.
- Keep ready the T&Ps, diesel driven dewatering pumps, diesel, torch lights, required PPEs and skilled manpower, for post cyclone restoration activities.
- After cyclone, identify the cyclone affected locations / damages, inform Head of Station and Head of O&M and initiate corrective actions.

(iii) Sweet Water Pump House Team:

The team will report to Head of O&M. *List of team members & their responsibilities as follows:*

1. Representative from OS	3. Representative from Operation
2. Representative from EM	4. Representative from Security

- Identify probable flooding areas and mobilize requisite curbing material, diesel driven pumps etc.
- Organize services like alternative power, battery operated towers lights, torch lights, etc for safe rehabilitation and restoration of pump house.
- During emergency, organize safe shutdown of pump house and evacuation of non-essential people, if necessary.
- Ensure isolation of power to probable drowning equipments.
- Keep ready sufficient stock of food items for essential staff and security personnel.

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- f. Immediately after cyclone, identify the cyclone affected locations / damages in the pump house, power supply lines and roads leading to the pump house, inform Head of Station and Head of O&M.
- g. In consultation with Head of O&M, attempt should be made for restoration of power, communication and access roads.

(iv) FM Team:

The team will report to Head of O&M. *List of team members & their responsibilities as follows:*

1. Head of FM	3. Head of CHP-M	2. Head of CHP-O
4. Head of CHP-EM	5. Head of FC	

- a. Identify probable flooding areas and mobilize requisite curbing material and diesel driven pumps at these locations.
- b. Organize services like alternative power, battery operated towers lights, torch lights to use in case of power failure.
- c. Lower the booms of all stacker/reclaimers, cranes, if any, and kept locked.
- d. Arrange to close doors & windows at all TPs and Crusher houses.
- e. Arrange to lower all lighting masts.
- f. Isolate power to the traction lines.
- g. Ensure the available rakes in the plant are appropriately positioned.
- h. During emergency, organize safe shutdown of CHP and evacuation of non-essential people from all locations of CHP.
- i. Call back all operators at all high rise TPs and Crusher house.
- j. Immediately after cyclone, identify the cyclone affected locations / damages at all locations of CHP including R&D cabin to TH/WT, traction lines and roads, and inform Head of Station and Head of O&M.
- k. In consultation with Head of O&M, attempt should be made for restoration of power, communication, access roads on priority basis.

(v) Off-sites Team:

The team will report to Head of O&M. *List of team members & their responsibilities as follows:*

1. Head of MM-OS	3. AGM(Operation)
2. Representative from EM	4. Representative from C&I

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- a. Identify probable flooding areas and mobilize requisite curbing material and diesel driven pumps at these locations.
- b. Organize services like alternative power, battery operated towers lights, torch lights to use in case of power failure.
- c. Arrange to tightly close gates, doors & windows at all pump houses.
- d. During emergency, organize safe shutdown of Pump houses and evacuation of non-essential people from all off-site locations.
- e. Immediately after cyclone, identify the cyclone affected locations / damages at pump houses and roads, and inform to Head of Station and Head of O&M.
- f. In consultation with Head of O&M, attempt should be made for restoration on priority basis.

(vi) Township Team:

The team will report to Head of O&M. *List of team members & their responsibilities as follows:*

1. Head of TAD	3. A representative from T/sh Civil
2. Head of Township-EM	4. A representative from T/sh EM

- a. Organize alternative power, battery operated towers lights, torch lights, tree cutting machines, sanitation material etc.
- b. Cyclone warning should be propagated by announcement thro PA System, where in the Residents should be informed about various precautions to be taken during cyclone. (ex. To not to wander on road, to park vehicles under parking sheds, to tightly close all doors, to not to open any door during storm, to keep in reserve water/food/milk/candles etc.).
- c. Arrange to lower lighting masts, luminary light weight boards, hoardings etc.
- d. Identify probable locations of trees falling on overhead electrical poles, transformers, sub- stations etc., and arrange to safeguard.
- e. During emergency, organize safe shutdown of power in the township.
- f. Immediately after cyclone, identify the cyclone affected locations / damages and inform to Head of Station and Head of O&M.
- g. In consultation with Head of the Station, attempt should be made for restoration

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on priority basis.

- h. Arrange suitable / competent manpower for clearing of roads from fallen trees.
- i. Organize with the help of CISF, restricting movement of vehicles & people on the township roads.

RESPONSIBILITIES OF RESCUE TEAM:

The rescue team would typically consist of personnel from NTPC and CISF-Fire wing as decided by Head of the Station and Head of O&M depending upon the impact of cyclone in the plant and township. This team would be assisted by CISF security personnel for handling the injured and also for rescue operation. Their responsibilities are;

1. Assess the situation and co-ordinate rescue operation such as evacuation of affected personnel.
2. Keep ready vehicles, ambulance, rescue material (ex. fire tender, trailer pump, stretchers, safety belts, PP rope, search lights, axe, hacksaw, tree cutting m/c, etc.)
3. Ensure availability of PPE and their safe use by the team members.
4. Ensure that sufficient numbers of trained personnel are available for rescue operations on site.
5. Immediately after the disaster, on the directions of CIC/IC, the team shall start search cum rescue operations at plant, pump houses and township on the order of priority.
6. Depending upon the requirement, the team shall separate in to two to three groups to carry out the search/rescue operations simultaneously.
7. The team leader shall communicate to CIC/IC from time to time, the operations anticipating the right direction.

RESPONSIBILITIES OF ADMINSTRATIVE TEAMS:

1. Administration Team (Food & Catering, Rehabilitation & Photo/videography):

The role of Food & Catering team is to provide the necessary relief aids during any

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disaster in the plant. The team and their responsibilities are as follows:

1. Head of HR	2. AGM(C&M)
3. AGM(Fin)	4. Representatives from HR

1. Plan for sufficient material for smooth catering of food and refreshments all the time during the emergency period.
2. Arrange for canteen services and sufficient no. of personnel on duty.
3. Organize the transportation of food & relief materials to all affected locations in the plant and pump house outside the plant.
4. Arrange for temporary shelters & manning for rehabilitation of those being evacuated from plant locations.
5. Arrange for head counts of employees, contractors, transporters and visitors.
6. Keep the employees informed in township and seek their help if necessary.
7. Keep ready photographers and videographers for coverage of emergency situation, impact of emergency and response activities.

2. Communication Team

The role of Communication team is to provide and ensure working of all types of communication systems and facilities in the plant. The Head of the team will be the head of IT Department. He will be assisted by his department personnel. *Their responsibilities are as follows.*

1. Maintaining the communication network in working condition during the period of emergency.
2. Attending urgent repairs in the communication system, if required.
3. Keeping ready the additional communication facilities like Walkie Talkies / Radios, etc for use in case of existing communication systems fail.
4. Remain in touch with CIC and WIC for instructions and guidance.

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3. Medical Team

The entire M&H services department personnel would be called as medical team. The following medical arrangements should be made ready by head of the team, i.e., CMO and his team. The assistance of trained first-aiders would be taken in handling the victims.

1. Keep ready stretchers, Ambulance, first aid equipment and trained first aid persons at Plant First aid Center and Hospital in the township.
2. Keep ready the additional no. of beds and medical infrastructure at hospital, ambulances, emergency vehicles for shifting of casualties, if any, required medicines.
3. Get in touch with CIC/WIC for any type of additional medical aid required and inform empaneled hospitals.
4. Ensure prioritization of patients/victims, administering medical help on priority.
5. Ensure first aid to the victims reporting directly at hospital without any paper formalities and delay.
6. Maintain records of affected persons, treatment given to them, etc.